

BrightLeain Exercise 1 SQL fundamentals

1. Retrieve all columns from the employees table
SQL

1. SELECT *
2. FROM employees;

Output

id	first_name	Last_name	department	Salary	hire_date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. Find all unique departments
SQL

- 1 SELECT DISTINCT department
- 2 FROM employees;

Output

department
IT
HR
Finance
Marketing

3. Retrieve first and last names ordered by descending SQL

- 1 SELECT first_name, last_name
- 2 FROM employees
- 3 ORDER BY salary DESC

Output

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees

SQL

- 1 SELECT id, first_name, last_name, salary
2. FROM employees
3. ORDER BY salary DESC
4. LIMIT 3

Output

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5. Find employees in the IT department

SQL

- 1 SELECT id, first-name, last-name, department
- 2 FROM employees
- 3 WHERE department = 'IT';

Output

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Find employees in Finance with Salary > 60000

SQL

- 1 SELECT id, first-name, last-name, department, Salary
- 2 FROM employees
- 3 WHERE department = 'Finance'
- 4 AND Salary > 60000;

Output

id	first-name	last-name	department	Salary
3	Cathy	Zulu	Finance	65000

7. Find employees in HR or Marketing

SQL

1. SELECT id, first-name, last-name, department
- 2 FROM employees
3. WHERE department = 'HR'
- 4 OR department = 'Marketing';

Output

Id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. find employees not in IT

SQL

1 SELECT id, first-name, last-name, department

2 FROM employees

3 WHERE ~~≠~~ NOT department = 'IT';

Output

Id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. find employees in IT, HR or Finance using IN

SQL

1 SELECT id, first-name, last-name, department

2 FROM employees

3 WHERE department IN ('IT', 'HR', 'Finance');

Output

Id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. Find employees in IT with salary > 65000 and city Johannesburg

SQL

- 1 SELECT id, first_name, last_name, department, salary, city
- 2 FROM employees
- 3 WHERE department = 'IT'
- 4 AND salary > 65000
- 5 AND city = 'Johannesburg'

Output

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg